

DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

 **Presentation:** Heike Thöricht

 **Bremen**, 23. August 2026

Agenda



The Challenge

Why demand forecasting matters



Our Approach

Data-driven solution



Model Performance

Proven accuracy



Key Insights

What drives demand



Implementation & Next Steps

How to use it

Executive Summary — Ready for Go-Live



Prediction Accuracy

95.7% of demand variance explained ($R^2 = 0.957$) — exceptional accuracy for confident planning



Forecast Error

±29 bikes on average (MAE = 29.1) — very tight margin for precise logistics planning



Key Driver

Historical demand patterns (lag_1: 36%) and time-of-day (18.1%) drive predictions — highly actionable

The model achieves exceptional accuracy using historical demand patterns and temporal features — production-ready to support your planning processes.

The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

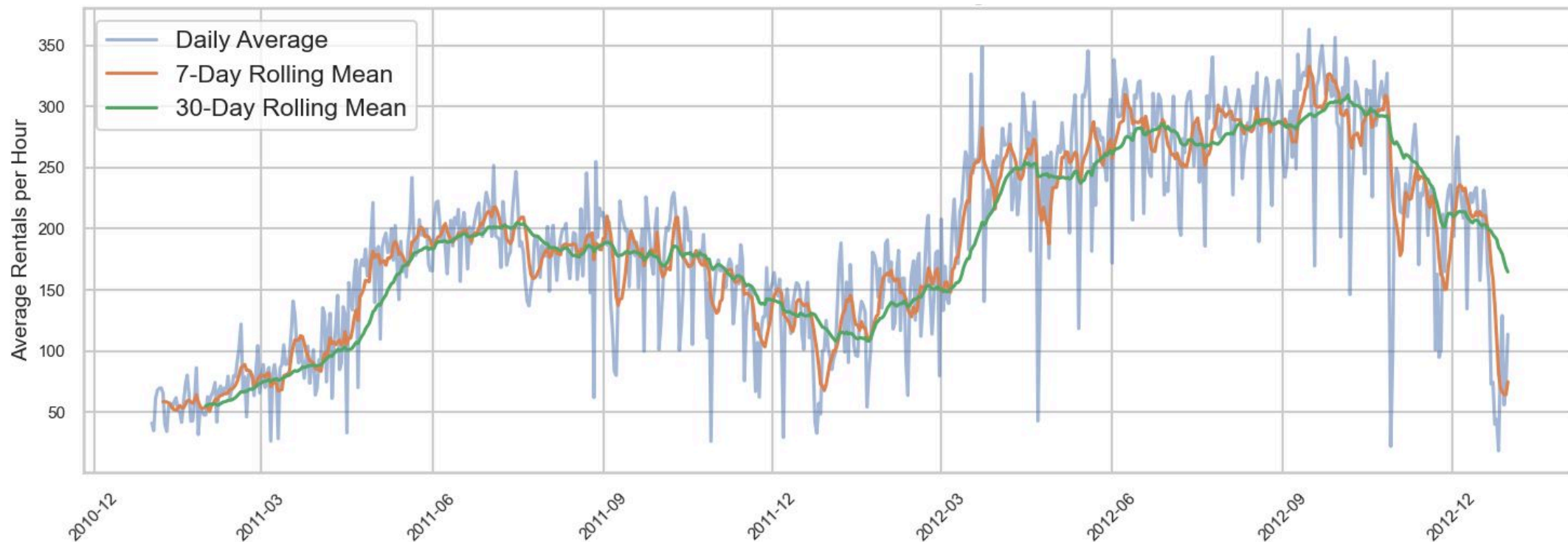
- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- Key insight: **Historical demand patterns (36%) + time-of-day (18%)** are the strongest predictors
- Goal: Accurate hourly forecasts based on **demand momentum and temporal patterns**

The Rise of Bike Sharing: Emerging Business Opportunities



Key Insight: Demand Patterns Drive Planning



Commuter Peaks

Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.



Leisure Patterns

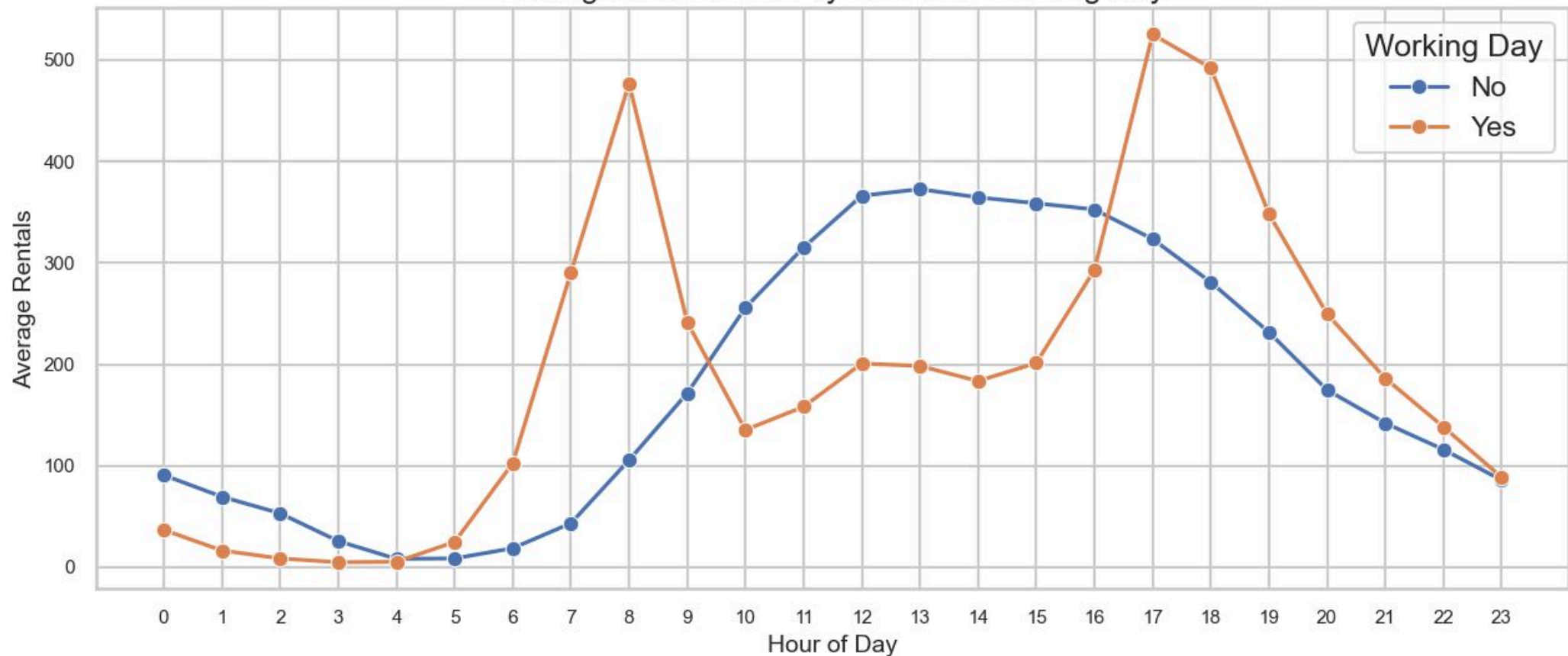
Non-working days show midday peaks — different staffing and bike distribution needed



Actionable

Hour-of-day is the **strongest predictor** — enables precise hourly planning

Average Bike Rentals by Hour and Working Day



These patterns are consistent and predictable — your planning can rely on them.

Model Performance: Proven Accuracy

Comparing our top three models across key performance metrics — XGBoost delivers **exceptional accuracy** with an R^2 of 0.957, making it the clear choice for production deployment.

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	V01	0.9033	47.99	68.56
Random Forest	V01	0.9547	28.99	46.91
 XGBoost	V01	0.9575	29.12	45.48

Winner: XGBoost

95.7% accuracy — explains nearly 96 out of 100 demand variations, delivering exceptional performance for production deployment

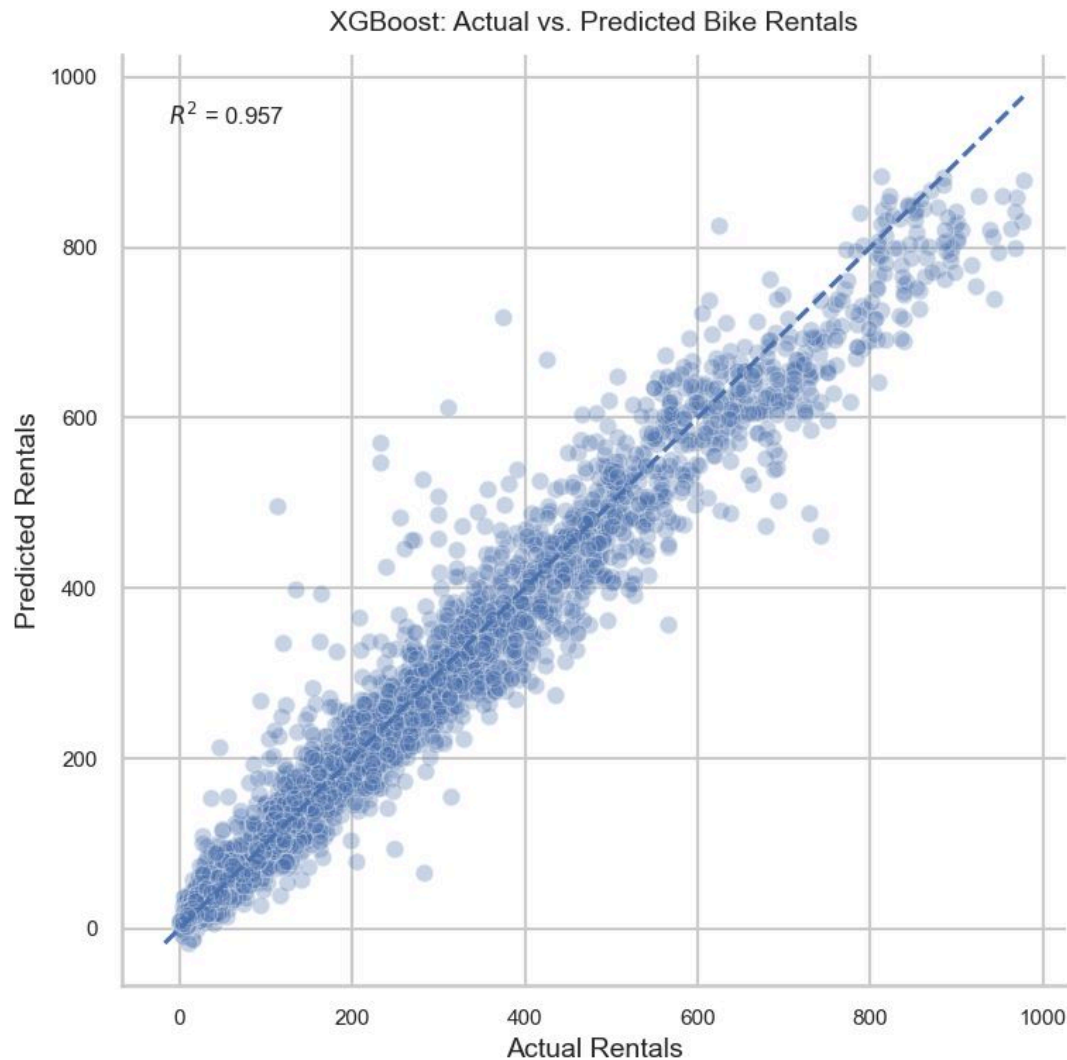
Forecast Error: ±29 Bikes

Average prediction error of just **29 rentals** — a very tight margin for operational planning and fleet distribution

Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios with exceptional accuracy

Validation: Model Works in Real Scenarios



✓ Accurate Predictions

Model predictions follow actual demand closely — **reliable for planning.**

⚠ Peak Hour Challenges

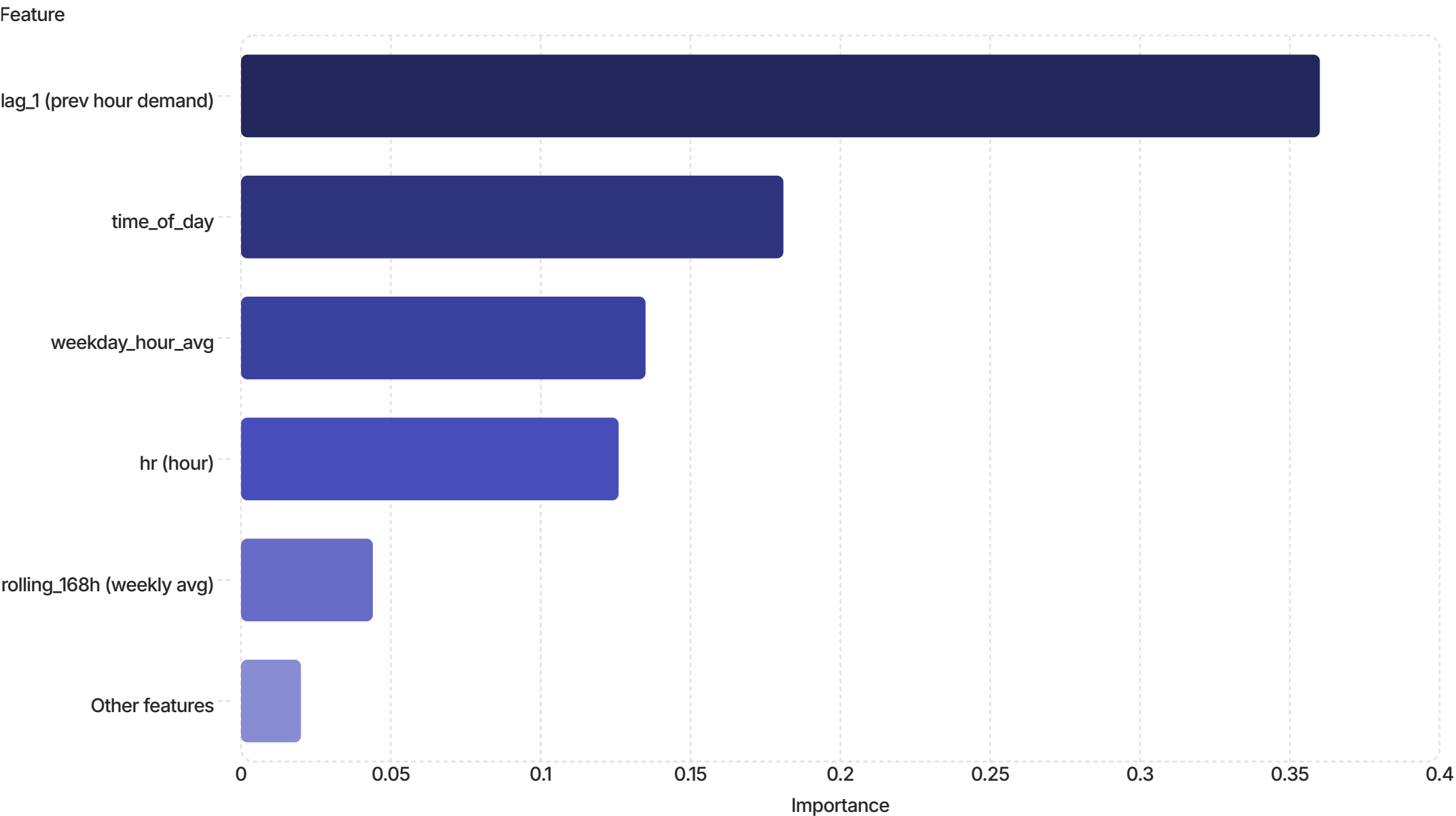
Errors slightly larger during peak hours — expected, but still **within acceptable range.**

🎯 Ready for Operations

Strong overall performance across all demand levels — **suitable for production use.**

The model is validated and ready to support your daily planning operations.

What Drives Demand: The Key Factors



Demand Momentum Dominates

Previous hour's demand (lag_1: 36%) is the strongest predictor — **demand patterns are highly persistent.**



Time Patterns Matter

Time-of-day (18.1%) and hour-of-day (12.6%) together account for **30.7%** — temporal patterns are critical.



Actionable

Focus on **recent demand trends + hourly patterns** — these drive 67% of predictions. Weather has minimal impact (<3%).

The model reveals the true drivers: demand momentum (36%) + temporal patterns (31%) together explain **67% of predictions.** Recent history is your best predictor.

Implementation: How to Use the Model



Forecast Refresh & Retraining

- **Daily:** Generate hourly forecasts each evening
- **Weekly:** Retrain model every Monday
- **Why:** lag_1 (36%) needs fresh data; weekday_hour_avg (13.5%) needs stable patterns

Owner: Data/Analytics



Go-Live Readiness

- Production-ready
- **Requires:** API integration, data pipeline, dashboard
- **Timeline:** 2–3 weeks to deployment

Owner: IT



Success Metrics

- **Target:** MAE < 50 bikes
- **Monitor:** Weekly forecast accuracy
- **Retrain:** Quarterly or when drift detected

Owner: Analytics

The model is ready. Assign owners, build the pipeline, and start forecasting from day one.

Questions & Next Steps

Ready to deploy. Let's discuss implementation timeline and resource allocation.

Heike Thöricht

Action Items



**Review forecast dashboard
prototype — *this week***



**Confirm API requirements with IT
— *next week***



**Schedule go-live meeting —
*within 2 weeks***